PROGRAM 3: Build a Vector Database Using FAISS and Perform Similarity-Based Retrieval

AIM

To create a vector database using FAISS and retrieve documents based on the similarity between their embeddings and a query embedding.

ALGORITHM

1. Install Sentence Transformers and FAISS.
2. Import the required libraries.
3. Prepare a collection of documents.
4. Generate document embeddings.
5. Normalize the embeddings.
6. Create a FAISS index.
7. Add the document vectors to the index.
8. Convert the search query into an embedding.
9. Search the index for the top three similar documents.
10. Display the retrieved documents and similarity scores.

CODE

Cell 1: Install the libraries

```
!pip -q install sentence-transformers faiss-cpu
```

Cell 2: Python program

```
import faiss

import numpy as np

from sentence_transformers import SentenceTransformer



# Load embedding model
```

```
model = SentenceTransformer("all-MiniLM-L6-v2")


# Documents
documents = [
    "Artificial intelligence is used in healthcare.",
    "Python is a popular programming language.",
    "Machine learning learns patterns from data.",
    "Cybersecurity protects computers and networks.",
    "Deep learning uses artificial neural networks."
]


# Generate embeddings
embeddings = model.encode(
    documents,
    convert_to_numpy=True
).astype("float32")


# Normalize vectors for cosine similarity
faiss.normalize_L2(embeddings)


# Create FAISS index
dimension = embeddings.shape[1]
index = faiss.IndexFlatIP(dimension)
```

```
# Store vectors in the index
index.add(embeddings)

# Search query
query = "How does AI learn patterns?"

query_vector = model.encode(
    [query],
    convert_to_numpy=True
).astype("float32")

faiss.normalize_L2(query_vector)

# Retrieve top 3 documents
scores, indices = index.search(query_vector, 3)

print("Query:", query)
print("\nRetrieved Documents:")

for score, idx in zip(scores[0], indices[0]):
    print(f"\nSimilarity Score: {score:.4f}")
    print(documents[idx])
```

INPUT

How does AI learn patterns?

OUTPUT

Query: How does AI learn patterns?

Retrieved Documents:

Similarity Score: [calculated score]

Machine learning learns patterns from data

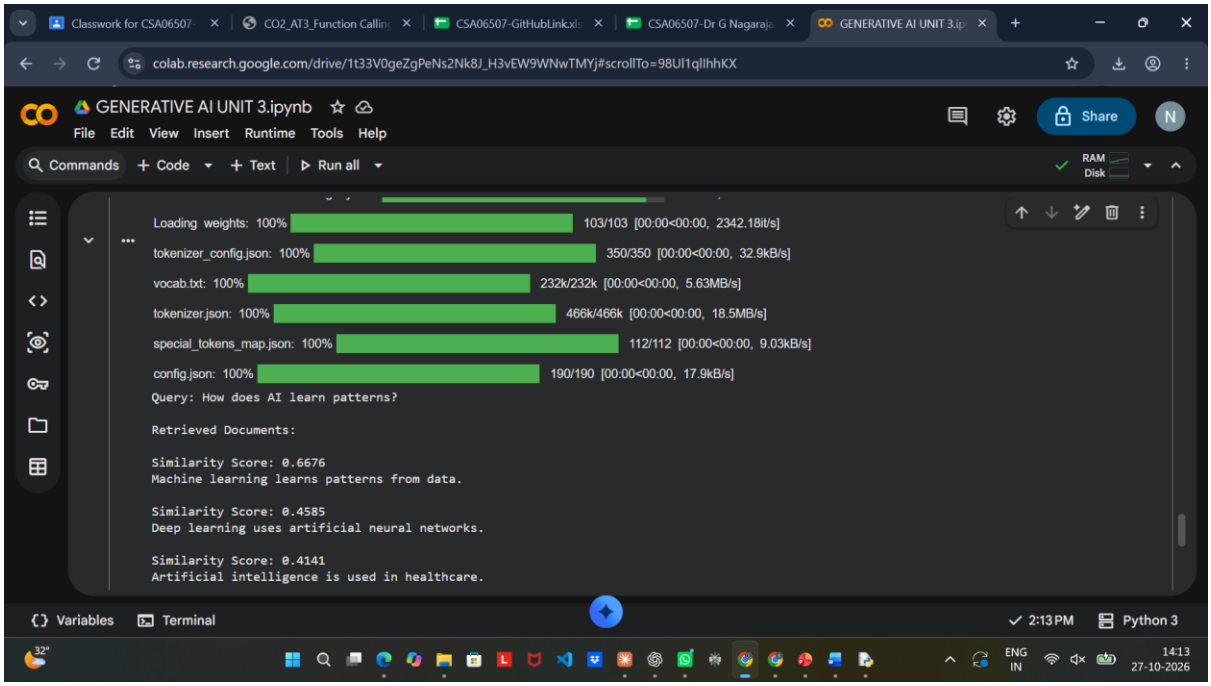
Similarity Score: [calculated score]

Deep learning uses artificial neural networks.

Similarity Score: [calculated score]

Artificial intelligence is used in healthcare.

Exact scores and ordering depend on the generated embeddings.



RESULT

A FAISS vector index was created successfully, and the most similar documents were retrieved using vector similarity search.